

Aditya

+91-9241551201 | kraditya9241@gmail.com | linkedin | github | adiresume

EDUCATION

Indian Institute of Technology, Patna

B.S. Computer Science and Data Analytics (CSDA)

Patna, Bihar

Mar. 2025 – Jun. 2029

EXPERIENCE

AI for Bharat

AI Engineer (Developer)

Patna, Bihar

Feb. 2026 – Mar. 2026

- Built **DataScout**, an agentic AI data analyst enabling natural-language data analysis using Amazon Bedrock and AWS serverless architecture (S3, Lambda, API Gateway, DynamoDB).
- Implemented LLM-generated Python code execution for analytical results, improving accuracy and auditability; developed Streamlit interface for no-SQL dataset insights.
- Architected enterprise-grade security: IAM role isolation, AES-256 encryption, audit logging, and sandboxed execution. Led a team of 4 in the Prototype Development Phase of a national AI hackathon.

WorldQuant BRAIN

Alpha Research Trainee

Remote

Jan. 2026 – Feb. 2026

- Designed and simulated equity alpha signals using WorldQuant BRAIN's Fast Expression language on the US TOP3000 stock universe with market and industry neutralization.
- Applied decay, truncation, and delay constraints to manage turnover and risk; evaluated performance using Sharpe ratio, turnover, drawdown, and out-of-sample test periods.

PROJECTS

Market Regime Detection Framework | Python, HMM, K-Means, XGBoost, MLflow, Streamlit

2025

- Authored an **IEEE-format research paper** proposing a hybrid framework combining Hidden Markov Models and K-Means clustering to identify 3 market regimes (low-vol, trending, crisis) on 15 years of NIFTY 50 data.
- Trained an XGBoost classifier with class-imbalance handling (sample weighting) achieving **78% regime prediction accuracy**; engineered 28 features including realized volatility, VIX dynamics, and return skewness.
- Dynamic strategy allocation (mean reversion / breakout / risk-off) achieved **14.2% CAGR vs. 10.8% buy-and-hold** with maximum drawdown reduced from 50% to 18%; successfully exited before 2008 and 2020 crashes.

DemandIQ – Retail Demand Forecasting & Replenishment Engine | Prophet, XGBoost, FastAPI, Streamlit

Mar. 2026

- Capstone-I project (IIT Patna): led model training for a 6-layer modular pipeline forecasting next-week sales per SKU per store using a dual-model approach (Prophet for seasonality, XGBoost for complex patterns).
- Engineered lag features (7/14/28-day), rolling statistics, price delta, and holiday flags; XGBoost outperformed Prophet on most SKUs based on MAE; tracked experiments with MLflow.
- Implemented safety stock formula ($Z \times \sigma \times \sqrt{\text{lead.time}}$) at 95% service level and real-time Telegram alerts for HIGH-risk inventory; served via FastAPI REST endpoints and Streamlit dashboard.

IndiEye – Market Intelligence Platform | Next.js, FastAPI, PostgreSQL, Llama 3, AWS

2026 – Present

- Founded and open-sourced an ongoing research startup democratizing institutional-grade market sentiment analysis for retail investors; actively developing and improving the platform.
- Built a real-time data ingestion pipeline aggregating financial news, social media chatter (X, Reddit), and daily SEC filings; fine-tuned Llama 3 for domain-specific sentiment extraction.
- Correlated NLP-derived sentiment signals with price action using time-series models; delivered insights through a predictive dashboard served via FastAPI backend on AWS with PostgreSQL storage.

LLM Pruning & Explainability Research | SHAP, Integrated Gradients, DistilBERT, RoBERTa

2026

- Investigated the impact of model compression (random, L1 unstructured, L1 structured pruning at 40–80% sparsity) on XAI faithfulness using FCor metric across DistilBERT and RoBERTa on IMDB and Yelp datasets.
- Found magnitude pruning preserves explanation faithfulness up to 80% sparsity; random pruning creates high-curvature decision landscapes (Hessian values up to 14,090 vs. 0.73 baseline) that break SHAP linearity assumptions.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, C++, Rust

ML / Quant: XGBoost, Prophet, HMM, K-Means, SHAP, Llama 3 (fine-tuning), MLflow, Backtesting, Alpha Research, Sharpe/Drawdown Analysis

Databases: PostgreSQL, DynamoDB

Web / APIs: FastAPI, Streamlit, Plotly, Next.js, React

Cloud / Infra: AWS (S3, Lambda, API Gateway, DynamoDB, Bedrock), IAM, AES-256 Encryption

Tools: Git, Pandas, NumPy, Scikit-learn, hmmlearn, yfinance, Uvicorn

Certifications: NISM Series VIII – Equity Derivatives Certification Examination